

# Block 11: complete runtime Solana schema v2 and bounded message signing

## 1. Identity and frozen scope

Owner: Codex /root, agent-1. Worktree: /private/tmp/kk715-stack11-agent1.  
Branch: audit/715-stack11-agent1-20260927. Target: BitHighlander/keepkey-firmware develop.  
Develop base: fc1e93746132553ad98ed60f4847c8d770732bf9. Report generation predecessor: c0a5ae01f90c66e92a9fbb0943f7cc31f63618de.  
Code/test qualification head: e631449c18f0f26755dc79f59ffea1d6863e8219. Completed block10 predecessor: 476afea2ce9906caaac67bffa900b6a961fa6c4.  
Python pin: aae89d378d889b3872696469fe363cf570e5ac4c. Final containing head and PR URL are recorded in the PR body.  
Main master-template SHA256: 4df8723670dbd7266ed8d8756b790725fb4f0f5886d36fe2f9b7c3aece2d64e0.  
Main SOP SHA256: cd2c546d41f0a723f6ce84199a3fb03661f79ab3f49fbafca08de96a7c309e89.

The owner's latest instruction requires the ENTIRE BLOCK PR against fork develop.  
This supersedes the older stacked-target instruction for this PR. Develop is an ancestor; the target diff necessarily includes accepted predecessor blocks as well as all block11 source. This is a cumulative integration PR, not a ten-line repair or a claim that earlier audits re-ran over the entire release. Predecessor evidence and outstanding release gates retain their original status. No merge is requested.

Block11 includes schema versions 1/2, eight v2 argument slots, token-amount and duration display, same-runtime-signer token metadata, exact instruction coverage, companion restrictions, bounded plain-text raw message signing, AdvancedMode and session trust, and ordinary consent. Certified roots, delegate certificates, LUT proofs and production publication are excluded from runtime 7.15 scope.

Historical PR841 head e97ecd49d03e49536a0fe690bea1319879db022b and base 58526221f47a06d44ef49c9a0642136423f6556f differ in four files despite the retirement comment saying tree-empty. Four core functions are retained from that head in the accepted predecessor; the intake JSON records exact comparisons. The full block's code is visible against develop, including its retained native tests. PR841 is historical; it is not reopened and its retirement is not evidence of an empty diff.

## 2. Git inventory and behavior coverage

Net target inventory command: git diff --numstat fc1e93746132553ad98ed60f4847c8d770732bf9 c0a5ae01f90c66e92a9fbb0943f7cc31f63618de.  
Every target path is listed in section 7. Counts include code, tests, dependencies, workflows, old receipts and this report. Historical counts overlap; do not sum them.

| Behavior                                         | Code-bearing paths in the develop-target PR                                                              | Evidence                                                                                    |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Schema v1/v2 bounded parser and widths           | include/keepkey/firmware/solana.h; lib/firmware/solana.c; unittests/firmware/solana.cpp                  | Native canonical/v2, unsafe-label, trailing-byte and all-prefix truncation assertions       |
| Exact program/discriminator/data/account binding | lib/firmware/solana.c; unittests/firmware/solana.cpp                                                     | Coverage/mismatch/index tests and transfer-companion wire refusal with valid control        |
| Runtime-only trust, token mint and same signer   | lib/firmware/fsm_msg_solana.h; lib/firmware/signed_metadata.c; deps/python-keepkey                       | Runtime token display control and eight invalid-attestation cases                           |
| Session reset and policy                         | lib/firmware/fsm_msg_common.h; lib/firmware/storage.c; deps/python-keepkey                               | Initialize/ClearSession/policy tests; newly added Solana Initialize refusal                 |
| Plain-text predicate and exact message consent   | lib/firmware/solana.c; lib/firmware/fsm_msg_solana.h; unittests/firmware/solana.cpp; deps/python-keepkey | Printable/control-byte/key-scan native tests and message-policy wire tests                  |
| Malformed input cannot reach schema consent      | lib/firmware/fsm_msg_solana.h; deps/python-keepkey                                                       | Three malformed forms, with/without schema; before/after wire evidence                      |
| Transport bounds and host consumer               | include/keepkey/transport/messages-solana.options; deps/python-keepkey                                   | raw_tx 2048, schema 256, signature 64, raw message 1024; client/protobuf consumer inspected |

| Behavior                   | Code-bearing paths in the develop-target PR | Evidence                                                     |
|----------------------------|---------------------------------------------|--------------------------------------------------------------|
| Accepted predecessor stack | Remaining cumulative paths in section 7     | Existing accepted reports included; not attributed to S11-01 |

Original unit inventories:

Commit a1b63ac5e75c6792f15f8d439e452eca2f2ecb45:

```
13 7 include/keepkey/firmware/solana.h
93 12 lib/firmware/fsm_msg_solana.h
32 7 lib/firmware/solana.c
36 0 unittests/firmware/solana.cpp
```

Commit a7daa79ded5c38afef37c23d02aea6fa671f28a9:

```
6 0 include/keepkey/firmware/solana.h
32 25 lib/firmware/fsm_msg_solana.h
13 0 lib/firmware/solana.c
58 0 unittests/firmware/solana.cpp
```

Dependency pins at qualification:

```
a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec code-signing-keys (heads/master)
8a392f70a5d5575ece3dfb35f115d4a4b27f497c deps/crypto/trezor-firmware (8a392f70a)
5fec9e6906a340be5eb3d795ec746769065b2db8 deps/device-protocol (dice-entropy-v1-83-g5fec9e6)
7888184f28509dba839e3683409443e0b5bb8948 deps/googletest (release-1.8.0-755-g7888184f)
aae89d378d889b3872696469fe363cf570e5ac4c deps/python-keepkey (v6.0.3-958-gaae89d3)
6dfbfdad5d9303ed190d1c3cb7bec34b565b6ce8 deps/qrenc/QR-Code-generator (v1.4.0-11-g6dfbfda)
71d356a1141624994cf613bd2d2583892e8e6d5a deps/sca-hardening/SecAESSTM32 (heads/keepkey)
```

### 3. Current-source reconciliation and findings

S11-01 FIXED: SolanaSignTx formerly evaluated a valid runtime schema after solana\_inspectTx returned MALFORMED with partially populated instructions. That branch preceded the final malformed rejection and skipped signer validation. The new early error zeroes the derived node and returns SyntaxError before any schema evaluation. Baseline emulator tests reached confirmation on all three schema-bearing cases; rejecting the prompt produced ActionCancelled. The fixed handler returns the malformed error with no confirmation for all six cases. The baseline test did not approve a signature; signing reachability is source tracing, not a claim of a captured baseline signature or network acceptance.

Security closure: the sensitive operation is Ed25519 signing with the device's derived private key; host input is raw\_tx plus a runtime-signed schema. The host observes protocol response and confirmation requests. Full firmware is affected; BTC does not dispatch Solana. During parsing/schema selection malformed payloads must yield no signing consent or signature. Positive controls are existing valid runtime-schema signing and token metadata reviews. Negative controls are legacy trailing bytes, missing v0 lookup count, and truncated lookup entry with/without schema. Baseline failures falsify the old guard order; fixed results prove early rejection. Artifacts are in the adjacent evidence archive and local receipt.

Parser exhaustion, safe labels, typed argument bounds and exact byte coverage were reviewed in current source. Runtime signer ID checks precede narrowing; signature verification requires exactly 64 bytes and AdvancedMode. Token display requires a matching mint, bounded decimals, valid symbol/signature and the same signer slot as the schema; otherwise it displays base units and mint. Companions cannot silently transfer value; priority fee disclosure remains in the schema review. Raw printable text rejects the actual signer key at every byte offset and retains message-byte consent. Other message formats retain AdvancedMode.

CI-TARGET-01 PENDING: develop has no KK\_RELEASE\_MISSING\_CAPABILITIES declaration. The inherited report validator uses the PR base as independent waiver authority, so the develop-target report gate cannot validate this staged release's ledger. The repository has no KK\_ACCEPTED\_WAIVER\_SHA variable for manual CI either. No waiver, gate or repository setting was changed to hide this incompatibility. This is a target/qualification blocker, not a newly found signing defect. It must be resolved before calling hosted qualification green or requesting review.

## 4. Verification and artifact identity

- Full integrated firmware-unit: 669 tests pass on e631449c18f0f26755dc79f59ffea1d6863e8219.
- Full emulator and firmware-unit builds pass with image 7438e53933d4.
- Solana/runtime/session wire suites: 39 pass, 14 subtests pass, three skip.
- Earlier S11-01 baseline: three schema-bearing negative cases fail; schema-free controls pass.
- Earlier fixed baseline: 44 native Solana tests and 34 wire tests pass.
- Block09/10 changes are merged, preserving Python and firmware predecessor content.
- Preflight status at report generation: Pending final frozen-head preflight; do not request review yet..
- Hosted CI: pending publication on the exact report-containing head; CI-TARGET-01 remains open.
- ARM/BTC: predecessor evidence exists; fresh integrated candidate qualification pending.
- Physical-device/OLED and power-cycle verification remain release gates.

Exact skip reasons:

- test\_relay\_certified\_v0\_no\_lookup\_proof\_reaches\_signer\_check: Firmware version 7.16.0 or higher is required to run this test
- test\_advanced\_mode\_is\_off\_after\_power\_cycle: power cycle needs an emulator process this harness owns; none is bound to udp/12144 (CI runs it as a separate container). Run locally, or record the unplug/replug as manual evidence.
- test\_signer\_dropped\_by\_power\_cycle: power cycle needs an emulator process this harness owns; none is bound to udp/12144 (CI runs it as a separate container). Run locally, or record the unplug/replug as manual evidence.

Resume checkpoint: original temporary worktree/logs were unavailable. The exact committed candidate was restored at /private/tmp/kk715-stack11-resume. All 11 archived artifact hashes verified; independent source review found no new code defect. The old provisional receipt is explicitly historical. Workflow report metadata now names Python PR122 instead of PR121. Runtime code/tests/pins remain identical to e631449c18f0f26755dc79f59ffea1d6863e8219. A fresh pre-push run is recorded in the PR body after the final report is frozen.

The smallest hosted qualification path needs separately authorized repository variable KK\_ACCEPTED\_WAIVER\_SHA=2483977523d9b6beab52a69429ace9aeefd00f76, the accepted block7b authority. Its seven-capability ledger contains all five candidate waivers. The existing manual workflow can then qualify this exact head with publishing disabled; no validator edit is needed. A successful manual run would be distinct from the incompatible develop-based PR event. Until approved, the setting remains absent and fresh hosted qualification remains pending.

## 5. Report render and local preflight

The PDF is generated from this Markdown source. Frozen final Git inventory and PDF legibility are checked before publication; the final receipt records those checks and the containing head. The entire target diff is code-bearing; all block11 behaviors map to visible implementation/test paths above. The Python regression is published as a companion branch and pinned in this candidate.

## 6. Copilot checkpoint and completion

Owner authorization: exactly one run. Requests used: 0. External review pending. No block11 completion or release acceptance claimed. Do not spend the run on a repair-only or receipt-only PR. CI-TARGET-01 must be reconciled before the request. All predecessor decisions and review caveats retain their original status.

## 7. Complete target diff inventory

| Added | Deleted | Path                                    |
|-------|---------|-----------------------------------------|
| 1     | 2       | .gitattributes                          |
| 755   | 587     | .github/workflows/ci.yml                |
| 8     | 8       | .github/workflows/mirror-base-image.yml |
| 275   | 69      | .github/workflows/release.yml           |
| 6     | 0       | .gitignore                              |

| Added | Deleted | Path                                                                     |
|-------|---------|--------------------------------------------------------------------------|
| 21    | 0       | .gitleaks.toml                                                           |
| 5     | 0       | .gitleaksignore                                                          |
| 3     | 3       | .gitmodules                                                              |
| 58    | 4       | CMakeLists.txt                                                           |
| 1     | 0       | cmake/caches/device.cmake                                                |
| 40    | 0       | cmake/toolchains/mingw-w64-x86_64.cmake                                  |
| 19    | 1       | deps/crypto/CMakeLists.txt                                               |
| 1     | 1       | deps/crypto/trezor-firmware                                              |
| 1     | 1       | deps/device-protocol                                                     |
| 1     | 1       | deps/python-keepkey                                                      |
| 192   | 0       | docs/DiceEntropy.md                                                      |
| 154   | 0       | docs/dice-vs-coldcard.md                                                 |
| 94    | 0       | docs/firmware/reviews/7.15.0-review-round2-remediation.md                |
| 124   | 0       | docs/release/7.14.3-COMBINED-CANDIDATE.md                                |
| 230   | 0       | docs/release/REHEARSAL-SOP.md                                            |
| 148   | 0       | docs/release/RELEASE-PROGRAM.md                                          |
| 63    | 0       | docs/release/audit-units/715-04-review-20260923.md                       |
| -     | -       | docs/release/audit-units/715-04-review-20260923.pdf                      |
| 61    | 0       | docs/release/audit-units/715-05-review-20260924.md                       |
| -     | -       | docs/release/audit-units/715-05-review-20260924.pdf                      |
| 678   | 0       | docs/release/audit-units/715-05-skips-20260924.tsv                       |
| 20    | 0       | docs/release/audit-units/715-06-evidence-sha256-20260924.txt             |
| 82    | 0       | docs/release/audit-units/715-06-history-20260924.tsv                     |
| -     | -       | docs/release/audit-units/715-06-intake-evidence-20260924.tgz             |
| 140   | 0       | docs/release/audit-units/715-06-reconciliation-20260924.md               |
| 185   | 0       | docs/release/audit-units/715-06-review-20260924.md                       |
| -     | -       | docs/release/audit-units/715-06-review-20260924.pdf                      |
| 663   | 0       | docs/release/audit-units/715-06-skips-20260924.tsv                       |
| -     | -       | docs/release/audit-units/715-06-validation-20260924.tgz                  |
| 1086  | 0       | docs/release/audit-units/715-06-validation-sha256-20260924.txt           |
| 144   | 0       | docs/release/audit-units/715-07-first-copilot-retrospective-20260924.md  |
| -     | -       | docs/release/audit-units/715-07-first-copilot-retrospective-20260924.pdf |
| 81    | 0       | docs/release/audit-units/715-07-full-audit-remediation-20260924.md       |
| -     | -       | docs/release/audit-units/715-07-full-audit-remediation-20260924.pdf      |
| 112   | 0       | docs/release/audit-units/715-07-history-20260924.tsv                     |
| 62    | 0       | docs/release/audit-units/715-07-intake-20260924.md                       |
| 141   | 0       | docs/release/audit-units/715-07-review-20260924.md                       |
| -     | -       | docs/release/audit-units/715-07-review-20260924.pdf                      |
| 577   | 0       | docs/release/audit-units/715-07-skips-20260924.tsv                       |
| -     | -       | docs/release/audit-units/715-07-validation-20260924.tgz                  |
| 112   | 0       | docs/release/audit-units/715-07-validation-sha256-20260924.txt           |

| Added | Deleted | Path                                                                  |
|-------|---------|-----------------------------------------------------------------------|
| 42    | 0       | docs/release/audit-units/715-07b-d-resplit-mapping-20260927.md        |
| 35    | 0       | docs/release/audit-units/715-07b-resplit-checks-20260927.json         |
| 74    | 0       | docs/release/audit-units/715-07b-resplit-receipt-20260927.md          |
| 128   | 0       | docs/release/audit-units/715-07b1-formatters-audit-20260926.md        |
| -     | -       | docs/release/audit-units/715-07b1-formatters-audit-20260926.pdf       |
| 119   | 0       | docs/release/audit-units/715-07b2a-groups-audit-20260927.md           |
| -     | -       | docs/release/audit-units/715-07b2a-groups-audit-20260927.pdf          |
| 121   | 0       | docs/release/audit-units/715-07b2b-iteration-audit-20260927.md        |
| -     | -       | docs/release/audit-units/715-07b2b-iteration-audit-20260927.pdf       |
| 136   | 0       | docs/release/audit-units/715-07b3r-embedded-audit-20260927.md         |
| -     | -       | docs/release/audit-units/715-07b3r-embedded-audit-20260927.pdf        |
| 203   | 0       | docs/release/audit-units/715-08-10-integration-20260927.md            |
| -     | -       | docs/release/audit-units/715-08-10-integration-20260927.pdf           |
| -     | -       | docs/release/audit-units/715-08-10-local-evidence-20260927.tgz        |
| 110   | 0       | docs/release/audit-units/715-08-audit-20260927.md                     |
| -     | -       | docs/release/audit-units/715-08-audit-20260927.pdf                    |
| 86    | 0       | docs/release/audit-units/715-08-consolidated-audit-20260927.md        |
| -     | -       | docs/release/audit-units/715-08-consolidated-audit-20260927.pdf       |
| -     | -       | docs/release/audit-units/715-08-local-evidence-20260927.tgz           |
| 175   | 0       | docs/release/audit-units/715-08-reconciliation-20260927.json          |
| 80    | 0       | docs/release/audit-units/715-08B-audit-20260927.md                    |
| -     | -       | docs/release/audit-units/715-08B-audit-20260927.pdf                   |
| 114   | 0       | docs/release/audit-units/715-09-audit-20260927.md                     |
| -     | -       | docs/release/audit-units/715-09-audit-20260927.pdf                    |
| -     | -       | docs/release/audit-units/715-09-local-evidence-20260927.tgz           |
| 145   | 0       | docs/release/audit-units/715-09-reconciliation-20260927.json          |
| 163   | 0       | docs/release/audit-units/715-10-audit-20260927.md                     |
| -     | -       | docs/release/audit-units/715-10-audit-20260927.pdf                    |
| 31    | 0       | docs/release/audit-units/715-10-completion-plan-20260927.md           |
| -     | -       | docs/release/audit-units/715-10-final-readiness-evidence-20260927.tgz |
| 73    | 0       | docs/release/audit-units/715-10-intake-20260927.md                    |
| 48    | 0       | docs/release/audit-units/715-10-reconciliation-20260927.json          |
| 539   | 0       | docs/release/audit-units/715-11-complete-block-audit-20260927.md      |
| -     | -       | docs/release/audit-units/715-11-complete-block-audit-20260927.pdf     |
| 200   | 0       | docs/release/audit-units/715-11-intake-20260927.json                  |
| 36    | 0       | docs/release/audit-units/715-11-intake-20260927.md                    |
| -     | -       | docs/release/audit-units/715-11-local-evidence-20260927.tgz           |
| 82    | 0       | docs/release/audit-units/715-11-malformed-schema.md                   |
| 26    | 0       | docs/release/audit-units/P01-002-artifact-routing.md                  |
| 27    | 0       | docs/release/audit-units/P01-003-firmware-packaging.md                |
| 12    | 0       | docs/release/audit-units/P01-004-report-inputs.md                     |

| Added | Deleted | Path                                                                                 |
|-------|---------|--------------------------------------------------------------------------------------|
| 16    | 0       | docs/release/audit-units/P01-008-compose-variant.md                                  |
| 27    | 0       | docs/release/audit-units/P02-001-tiny-lifetime.md                                    |
| 14    | 0       | docs/release/audit-units/P02-003-packet-lifetime.md                                  |
| 18    | 0       | docs/release/audit-units/P06-002-ripple-display-response.md                          |
| 15    | 0       | docs/release/audit-units/P06-003-ripple-display-test.md                              |
| 21    | 0       | docs/release/audit-units/P06-005-ripple-unsupported-memo.md                          |
| 12    | 0       | docs/release/audit-units/P06-006-ripple-memo-rejection-test.md                       |
| 23    | 0       | docs/release/audit-units/decode-boundaries.md                                        |
| 15    | 0       | docs/release/audit-units/eos-authorization.md                                        |
| 87    | 0       | docs/release/audit-units/full-diff-audit-20260912.md                                 |
| 14    | 0       | docs/release/audit-units/passphrase-transition.md                                    |
| 32    | 0       | docs/release/audit-units/scope-repair.md                                             |
| 14    | 0       | docs/release/audit-units/source-storage-policy.md                                    |
| 21    | 0       | docs/release/audit-units/storage-capacities.md                                       |
| 11    | 0       | docs/release/audit-units/storage-cipher-cleanup.md                                   |
| 11    | 0       | docs/release/audit-units/storage-revocation.md                                       |
| 6     | 0       | docs/release/audit-units/token-build-dependencies.md                                 |
| 170   | 0       | docs/security/7.14.3-bitcoin-only-dice-audit-sop.md                                  |
| 235   | 0       | docs/security/HANDOFF-ERC7730-715-FORMATTERS.md                                      |
| 295   | 0       | docs/security/HANDOFF-ERC7730-ALPHA.md                                               |
| 209   | 0       | docs/security/HANDOFF-ERC7730-PHASE-E.md                                             |
| -     | -       | docs/security/evidence/rom-printf-integer-percent/01-thorchain-withdraw-25.05pct.png |
| -     | -       | docs/security/evidence/rom-printf-integer-percent/02-thorchain-sending-eth.png       |
| 17    | 0       | docs/security/evidence/rom-printf-integer-percent/README.md                          |
| 28    | 0       | include/keepkey/board/bsd_compat.h                                                   |
| 67    | 1       | include/keepkey/board/confirm_sm.h                                                   |
| 35    | 0       | include/keepkey/board/draw.h                                                         |
| 16    | 0       | include/keepkey/board/keepkey_display.h                                              |
| 21    | 0       | include/keepkey/board/layout.h                                                       |
| 57    | 24      | include/keepkey/board/messages.h                                                     |
| 5     | 0       | include/keepkey/board/usb.h                                                          |
| 61    | 17      | include/keepkey/emulator/libkemu.h                                                   |
| 43    | 1       | include/keepkey/firmware/app_confirm.h                                               |
| 9     | 1       | include/keepkey/firmware/app_layout.h                                                |
| 11    | 1       | include/keepkey/firmware/authenticator.h                                             |
| 12    | 0       | include/keepkey/firmware/binance.h                                                   |
| 22    | 0       | include/keepkey/firmware/bip85.h                                                     |
| 2     | 0       | include/keepkey/firmware/coins.def                                                   |
| 3     | 1       | include/keepkey/firmware/coins.h                                                     |
| 72    | 0       | include/keepkey/firmware/dice_input.h                                                |
| 11    | 4       | include/keepkey/firmware/eip712.h                                                    |

| Added | Deleted | Path                                                     |
|-------|---------|----------------------------------------------------------|
| 227   | 0       | include/keepkey/firmware/eip712_stream.h                 |
| 4     | 0       | include/keepkey/firmware/eos.h                           |
| 80    | 0       | include/keepkey/firmware/erc7730_abi.h                   |
| 90    | 0       | include/keepkey/firmware/erc7730_abi_stream.h            |
| 111   | 0       | include/keepkey/firmware/erc7730_capabilities.h          |
| 241   | 0       | include/keepkey/firmware/erc7730_catalog.h               |
| 20    | 0       | include/keepkey/firmware/erc7730_condition.h             |
| 69    | 0       | include/keepkey/firmware/erc7730_field.h                 |
| 38    | 0       | include/keepkey/firmware/erc7730_format.h                |
| 311   | 0       | include/keepkey/firmware/erc7730_program.h               |
| 26    | 0       | include/keepkey/firmware/erc7730_tx.h                    |
| 277   | 0       | include/keepkey/firmware/erc7730_workflow.h              |
| 20    | 2       | include/keepkey/firmware/ethereum.h                      |
| 34    | 0       | include/keepkey/firmware/ethereum_contracts.h            |
| 3     | 0       | include/keepkey/firmware/ethereum_contracts/saproxy.h    |
| 27    | 1       | include/keepkey/firmware/ethereum_contracts/thortx.h     |
| 3     | 0       | include/keepkey/firmware/ethereum_contracts/zxliquidtx.h |
| 14    | 3       | include/keepkey/firmware/ethereum_tokens.h               |
| 38    | 0       | include/keepkey/firmware/fsm.h                           |
| 98    | 0       | include/keepkey/firmware/hive.h                          |
| 2     | 0       | include/keepkey/firmware/home_sm.h                       |
| 54    | 4       | include/keepkey/firmware/mayachain.h                     |
| 36    | 1       | include/keepkey/firmware/osmosis.h                       |
| 7     | 0       | include/keepkey/firmware/pin_sm.h                        |
| 9     | 0       | include/keepkey/firmware/recovery_cipher.h               |
| 65    | 4       | include/keepkey/firmware/reset.h                         |
| 28    | 2       | include/keepkey/firmware/ripple.h                        |
| 242   | 0       | include/keepkey/firmware/signed_metadata.h               |
| 21    | 1       | include/keepkey/firmware/signing.h                       |
| 16    | 3       | include/keepkey/firmware/signtx_tendermint.h             |
| 148   | 0       | include/keepkey/firmware/solana.h                        |
| 45    | 1       | include/keepkey/firmware/storage.h                       |
| 30    | 0       | include/keepkey/firmware/tendermint.h                    |
| 50    | 5       | include/keepkey/firmware/thorchain.h                     |
| 4     | 1       | include/keepkey/firmware/tiny-json.h                     |
| 0     | 7       | include/keepkey/firmware/tokens.def                      |
| 10    | 1       | include/keepkey/firmware/transaction.h                   |
| 56    | 0       | include/keepkey/firmware/tron.h                          |
| 3     | 1       | include/keepkey/firmware/txin_check.h                    |
| 431   | 0       | include/keepkey/firmware/zcash.h                         |
| 19    | 0       | include/keepkey/rand/rng.h                               |

| Added | Deleted | Path                                                 |
|-------|---------|------------------------------------------------------|
| 143   | 0       | include/keepkey/rand/rng_health.h                    |
| 2     | 0       | include/keepkey/transport/interface.h                |
| 27    | 0       | include/keepkey/transport/messages-ethereum.options  |
| 56    | 0       | include/keepkey/transport/messages-hive.options      |
| 0     | 1       | include/keepkey/transport/messages-osmosis.options   |
| 2     | 0       | include/keepkey/transport/messages-ripple.options    |
| 12    | 0       | include/keepkey/transport/messages-solana.options    |
| 1     | 0       | include/keepkey/transport/messages-thorchain.options |
| 54    | 0       | include/keepkey/transport/messages-zcash.options     |
| 16    | 0       | include/keepkey/transport/messages.options           |
| 222   | 113     | include/pb.h                                         |
| 512   | 30      | lib/board/confirm_sm.c                               |
| 163   | 16      | lib/board/draw.c                                     |
| 4     | 1       | lib/board/font.c                                     |
| 13    | 10      | lib/board/keepkey_board.c                            |
| 16    | 6       | lib/board/keepkey_flash.c                            |
| 152   | 9       | lib/board/layout.c                                   |
| 162   | 68      | lib/board/messages.c                                 |
| 39    | 9       | lib/board/signatures.c                               |
| 28    | 2       | lib/board/timer.c                                    |
| 3     | 0       | lib/board/udp.c                                      |
| 50    | 23      | lib/board/usb.c                                      |
| 48    | 16      | lib/board/util.c                                     |
| 7     | 0       | lib/emulator/CMakeLists.txt                          |
| 302   | 35      | lib/emulator/libkemu.c                               |
| 45    | 5       | lib/emulator/setup.c                                 |
| 8     | 2       | lib/emulator/udp.c                                   |
| 56    | 27      | lib/firmware/CMakeLists.txt                          |
| 281   | 44      | lib/firmware/app_confirm.c                           |
| 117   | 4       | lib/firmware/app_layout.c                            |
| 240   | 131     | lib/firmware/authenticator.c                         |
| 141   | 14      | lib/firmware/binance.c                               |
| 105   | 0       | lib/firmware/bip85.c                                 |
| 19    | 1       | lib/firmware/coins.c                                 |
| 448   | 0       | lib/firmware/dice_input.c                            |
| 460   | 255     | lib/firmware/eip712.c                                |
| 1456  | 0       | lib/firmware/eip712_stream.c                         |
| 21    | 8       | lib/firmware/eos-contracts/eosio.system.c            |
| 117   | 13      | lib/firmware/eos.c                                   |
| 472   | 0       | lib/firmware/erc7730_abi.c                           |
| 463   | 0       | lib/firmware/erc7730_abi_stream.c                    |

| Added | Deleted | Path                                           |
|-------|---------|------------------------------------------------|
| 243   | 0       | lib/firmware/erc7730_capabilities.c            |
| 1652  | 0       | lib/firmware/erc7730_catalog.c                 |
| 109   | 0       | lib/firmware/erc7730_condition.c               |
| 297   | 0       | lib/firmware/erc7730_field.c                   |
| 233   | 0       | lib/firmware/erc7730_format.c                  |
| 877   | 0       | lib/firmware/erc7730_program.c                 |
| 46    | 0       | lib/firmware/erc7730_tx.c                      |
| 1097  | 0       | lib/firmware/erc7730_workflow.c                |
| 659   | 119     | lib/firmware/ethereum.c                        |
| 51    | 11      | lib/firmware/ethereum_contracts.c              |
| 36    | 24      | lib/firmware/ethereum_contracts/makerdao.c     |
| 52    | 13      | lib/firmware/ethereum_contracts/saproxy.c      |
| 189   | 47      | lib/firmware/ethereum_contracts/thortx.c       |
| 113   | 83      | lib/firmware/ethereum_contracts/zxapliquid.c   |
| 154   | 111     | lib/firmware/ethereum_contracts/zxliquidtx.c   |
| 201   | 36      | lib/firmware/ethereum_contracts/zxswap.c       |
| 80    | 16      | lib/firmware/ethereum_contracts/zxtransERC20.c |
| 21    | 13      | lib/firmware/ethereum_tokens.c                 |
| 416   | 19      | lib/firmware/fsm.c                             |
| 58    | 17      | lib/firmware/fsm_msg_binance.h                 |
| 143   | 0       | lib/firmware/fsm_msg_bip85.h                   |
| 52    | 9       | lib/firmware/fsm_msg_coin.h                    |
| 143   | 35      | lib/firmware/fsm_msg_common.h                  |
| 118   | 23      | lib/firmware/fsm_msg_cosmos.h                  |
| 27    | 14      | lib/firmware/fsm_msg_crypto.h                  |
| 24    | 7       | lib/firmware/fsm_msg_debug.h                   |
| 22    | 3       | lib/firmware/fsm_msg_eos.h                     |
| 2239  | 141     | lib/firmware/fsm_msg_ethereum.h                |
| 491   | 0       | lib/firmware/fsm_msg_hive.h                    |
| 162   | 30      | lib/firmware/fsm_msg_mayachain.h               |
| 210   | 113     | lib/firmware/fsm_msg_osmosis.h                 |
| 81    | 9       | lib/firmware/fsm_msg_ripple.h                  |
| 814   | 153     | lib/firmware/fsm_msg_solana.h                  |
| 125   | 23      | lib/firmware/fsm_msg_tendermint.h              |
| 136   | 29      | lib/firmware/fsm_msg_thorchain.h               |
| 37    | 55      | lib/firmware/fsm_msg_ton.h                     |
| 111   | 15      | lib/firmware/fsm_msg_tron.h                    |
| 1699  | 0       | lib/firmware/fsm_msg_zcash.h                   |
| 439   | 0       | lib/firmware/hive.c                            |
| 46    | 8       | lib/firmware/home_sm.c                         |
| 389   | 97      | lib/firmware/mayachain.c                       |

| Added | Deleted | Path                                             |
|-------|---------|--------------------------------------------------|
| 54    | 0       | lib/firmware/messagemap.def                      |
| 5     | 2       | lib/firmware/nano.c                              |
| 270   | 37      | lib/firmware/osmosis.c                           |
| 77    | 6       | lib/firmware/passphrase_sm.c                     |
| 47    | 21      | lib/firmware/pin_sm.c                            |
| 213   | 90      | lib/firmware/recovery_cipher.c                   |
| 482   | 96      | lib/firmware/reset.c                             |
| 65    | 10      | lib/firmware/ripple.c                            |
| 1046  | 0       | lib/firmware/signed_metadata.c                   |
| 549   | 59      | lib/firmware/signing.c                           |
| 193   | 65      | lib/firmware/signtx_tendermint.c                 |
| 538   | 41      | lib/firmware/solana.c                            |
| 306   | 42      | lib/firmware/storage.c                           |
| 8     | 0       | lib/firmware/storage.h                           |
| 180   | 4       | lib/firmware/tendermint.c                        |
| 318   | 46      | lib/firmware/thorchain.c                         |
| 5     | 3       | lib/firmware/tiny-json.c                         |
| 5     | 3       | lib/firmware/ton.c                               |
| 180   | 18      | lib/firmware/transaction.c                       |
| 364   | 1       | lib/firmware/tron.c                              |
| 14    | 7       | lib/firmware/txin_check.c                        |
| 11    | 2       | lib/firmware/u2f.c                               |
| 1237  | 0       | lib/firmware/zcash.c                             |
| 2     | 1       | lib/rand/CMakeLists.txt                          |
| 97    | 13      | lib/rand/rng.c                                   |
| 348   | 0       | lib/rand/rng_health.c                            |
| 18    | 0       | lib/transport/CMakeLists.txt                     |
| 2     | 1       | lib/transport/pb_decode.c                        |
| 1     | 1       | scripts/build/docker/device/debug.sh             |
| 7     | 2       | scripts/build/docker/device/release.sh           |
| 1     | 1       | scripts/build/docker/emulator/debug.sh           |
| 1     | 0       | scripts/cppcheck-version                         |
| 32    | 0       | scripts/cppcheck.sh                              |
| 10    | 2       | scripts/emulator/Dockerfile                      |
| 111   | 0       | scripts/emulator/capture-dice-flow.py            |
| 99    | 0       | scripts/emulator/capture-thor-percent.py         |
| 12    | 0       | scripts/emulator/docker-compose.bitcoin-only.yml |
| 14    | 0       | scripts/emulator/docker-compose.yml              |
| 19    | 1       | scripts/emulator/firmware-unit.sh                |
| 87    | 65      | scripts/emulator/python-keepkey-tests.sh         |
| 1     | 1       | scripts/emulator/python-keepkey.Dockerfile       |

| Added | Deleted | Path                                         |
|-------|---------|----------------------------------------------|
| 443   | 0       | scripts/emulator/test_stack07_regressions.py |
| 36    | 0       | scripts/emulator/test_stack09_integration.py |
| 141   | 0       | scripts/emulator/test_stack10_regressions.py |
| 623   | 68      | scripts/generate-test-report.py              |
| 103   | 0       | scripts/preflight.sh                         |
| 92    | 0       | scripts/preflight_checks.py                  |
| 248   | 0       | scripts/test_generate_test_report.py         |
| 124   | 0       | scripts/test_preflight_checks.py             |
| 91    | 0       | scripts/test_preflight_cppcheck.py           |
| 85    | 0       | scripts/tests/test_usb_receive_callbacks.py  |
| 36    | 0       | scripts/verify-token-def.py                  |
| 1     | 0       | tools/CMakeLists.txt                         |
| 124   | 0       | tools/check_sram_budget.py                   |
| 33    | 11      | tools/emulator/CMakeLists.txt                |
| 20    | 0       | tools/erc7730-validate/CMakeLists.txt        |
| 76    | 0       | tools/erc7730-validate/main.c                |
| 3     | 1       | tools/firmware/CMakeLists.txt                |
| 10    | 0       | tools/firmware/keepkey.ld                    |
| 14    | 9       | tools/firmware/keepkey_main.c                |
| 10    | 0       | tools/sram-budgets.json                      |
| 251   | 0       | tools/verify_dice_seed.py                    |
| 1     | 1       | unittests/board/CMakeLists.txt               |
| 490   | 2       | unittests/board/board.cpp                    |
| 17    | 2       | unittests/crypto/CMakeLists.txt              |
| 500   | 0       | unittests/crypto/bip340.cpp                  |
| 21    | 0       | unittests/crypto/emulator_random.c           |
| 55    | 7       | unittests/firmware/CMakeLists.txt            |
| 48    | 0       | unittests/firmware/app_confirm.cpp           |
| 302   | 0       | unittests/firmware/authenticator.cpp         |
| 151   | 0       | unittests/firmware/binance.cpp               |
| 28    | 2       | unittests/firmware/coins.cpp                 |
| 141   | 0       | unittests/firmware/confirm_test_utils.cpp    |
| 166   | 8       | unittests/firmware/cosmos.cpp                |
| 116   | 0       | unittests/firmware/dice.cpp                  |
| 194   | 0       | unittests/firmware/eip712.cpp                |
| 1076  | 0       | unittests/firmware/eip712_stream.cpp         |
| 90    | 3       | unittests/firmware/eos.cpp                   |
| 50    | 0       | unittests/firmware/eos_authorization.cpp     |
| 16    | 0       | unittests/firmware/eos_authorization_probe.c |
| 174   | 0       | unittests/firmware/erc7730_abi.cpp           |
| 371   | 0       | unittests/firmware/erc7730_abi_stream.cpp    |

| Added | Deleted | Path                                      |
|-------|---------|-------------------------------------------|
| 1901  | 0       | unittests/firmware/erc7730_catalog.cpp    |
| 105   | 0       | unittests/firmware/erc7730_condition.cpp  |
| 304   | 0       | unittests/firmware/erc7730_field.cpp      |
| 224   | 0       | unittests/firmware/erc7730_format.cpp     |
| 411   | 0       | unittests/firmware/erc7730_program.cpp    |
| 74    | 0       | unittests/firmware/erc7730_tx.cpp         |
| 207   | 0       | unittests/firmware/erc7730_workflow.cpp   |
| 709   | 1       | unittests/firmware/ethereum.cpp           |
| 1204  | 0       | unittests/firmware/fsm.cpp                |
| 783   | 0       | unittests/firmware/hive.cpp               |
| 20    | 0       | unittests/firmware/kkconfirm_driver.h     |
| 54    | 0       | unittests/firmware/liquidity_key_probe.c  |
| 390   | 7       | unittests/firmware/mayachain.cpp          |
| 11    | 0       | unittests/firmware/messages_probe.c       |
| 35    | 0       | unittests/firmware/nanopb_bounds.cpp      |
| 153   | 0       | unittests/firmware/osmosis.cpp            |
| 46    | 0       | unittests/firmware/recovery.cpp           |
| 37    | 0       | unittests/firmware/ripple.cpp             |
| 297   | 0       | unittests/firmware/rng_health.cpp         |
| 193   | 0       | unittests/firmware/setup_ceremony.cpp     |
| 1804  | 0       | unittests/firmware/signed_metadata.cpp    |
| 143   | 0       | unittests/firmware/signing.cpp            |
| 969   | 15      | unittests/firmware/solana.cpp             |
| 316   | 29      | unittests/firmware/storage.cpp            |
| 129   | 0       | unittests/firmware/storage_cipher_probe.c |
| 24    | 0       | unittests/firmware/storage_cipher_probe.h |
| 140   | 0       | unittests/firmware/storage_passphrase.cpp |
| 28    | 0       | unittests/firmware/test_board.cpp         |
| 350   | 21      | unittests/firmware/thorchain.cpp          |
| 173   | 0       | unittests/firmware/transaction.cpp        |
| 604   | 0       | unittests/firmware/tron.cpp               |
| 169   | 0       | unittests/firmware/usb_rx.cpp             |
| 2204  | 0       | unittests/firmware/zcash.cpp              |